

TERM PROJECT FINAL REPORT

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10 Nov 2025.

The report provides the progress on the project “Post-combustion CO₂ capture from coal combustion flue gas using solvent absorption and stripper-regenerator system“ and it covers a detailed flowchart of the system with streams labelled. The report identifies all given data, clearly distinguishes the unknowns to be solved and states the necessary assumptions to be made to setup the problem.

The Final report provides the complete calculations and analysis for the term project; it has the values upon calculations for all process units and provides the analysis for carbon capture. It also contains calculations for absorber sizing.

Background and motivation- With global pressure to decarbonize, thermal power plants face increasing scrutiny as they remain major contributors to CO₂ emissions. Carbon capture and storage (CCS) technologies, particularly post-combustion absorption using solvents, are emerging as practical retrofitting options for existing plants. Solvents as MEA,MDEA and NaOH are CO₂ absorbers.

Problem Statement- The problem selected was coal combustion and CO₂ capture wherein CO₂ was captured using the MEA solvent and absorber sizing was performed. The composition for coal was provided which helped in calculation for flue gas composition relevant data required was provided and then calculations for absorber column were supposed to be performed,analysis for use of MEA over other solvent and sustainability angles were to discussed relevance of purge was also to be discussed.

Basis of calculations

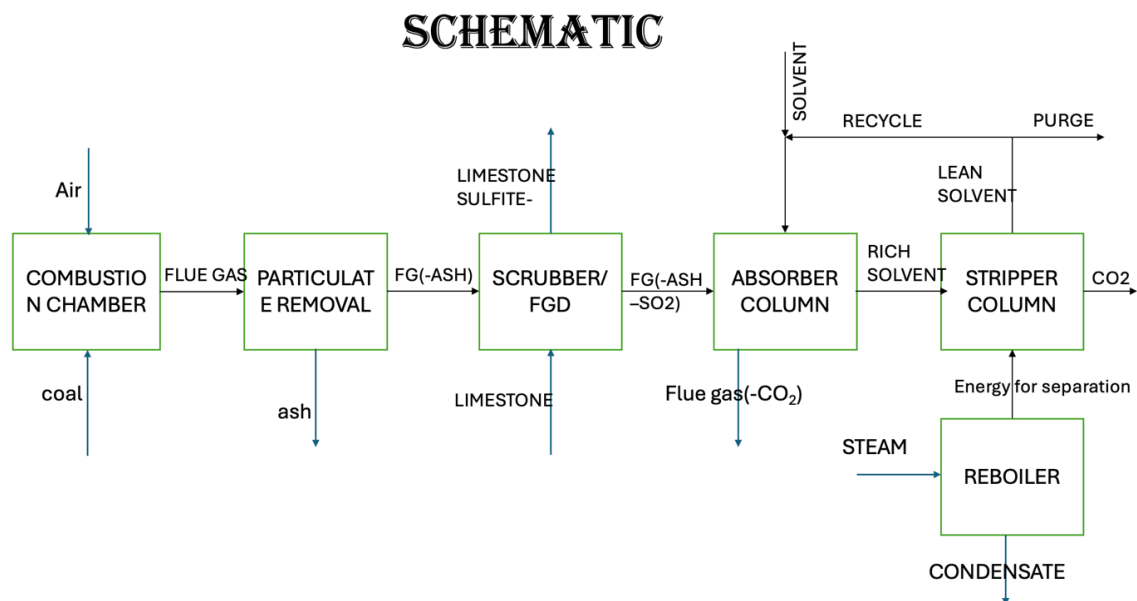
All calculations in this problem are performed on a basis of 10000 kg/h of coal(dry basis) feed to the combustor. The system is considered at a steady state with atmospheric pressure in the absorber and stripper columns unless specified.

Known values and general assumptions

- Ultimate analysis of coal gives us : C=48.10% H=3.10% N=1.40% S=0.60% O=8.56% Ash=38.24%.(all percentages are weight percentages).
- Complete combustion is assumed all C to CO₂, S to SO₂, H to H₂O .
- Air composition is 21% oxygen and 79%nitrogen by volume.

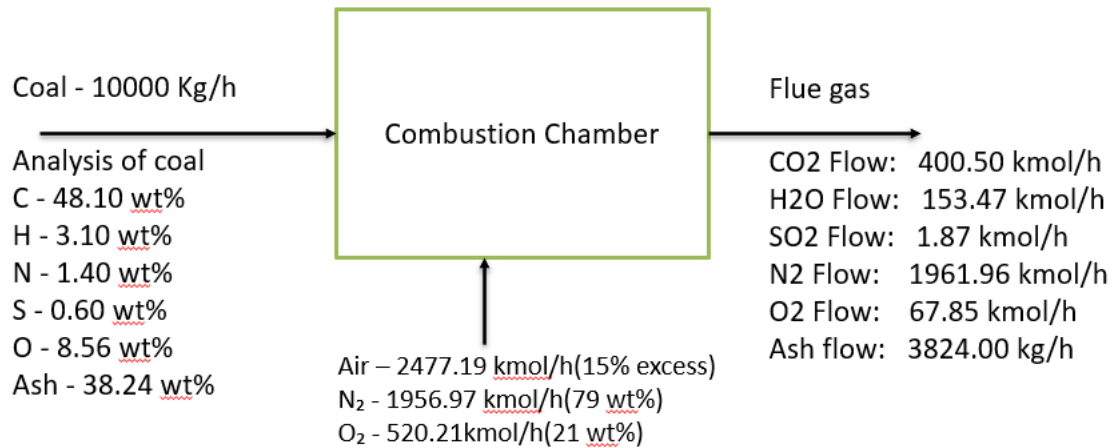
- Excess air is 15%.
- Target CO₂ capture is 90% of CO₂ generated.
- Solvent is 30 wt% MEA aqueous solution lean solvent loading is 0.2mol CO₂/mol MEA and rich solvent loading is 0.5mol CO₂/mol MEA.
- Pressure drops in absorber and stripper is neglected.
- Heat losses to surroundings are assumed negligible and hence neglected.
- FGD efficiency is 95% removal of SO₂.
- The superficial gas velocity is 2m/s in the absorber column.
- Reboiler duty is taken as 3.5MJ/kg CO₂.
- H of steam used to provide energy to the stripper is 2257KJ/Kg at 1 atm standard steam.
- Complete operation is considered to work under steady-state.

COMPLETE FLOW DIAGRAM

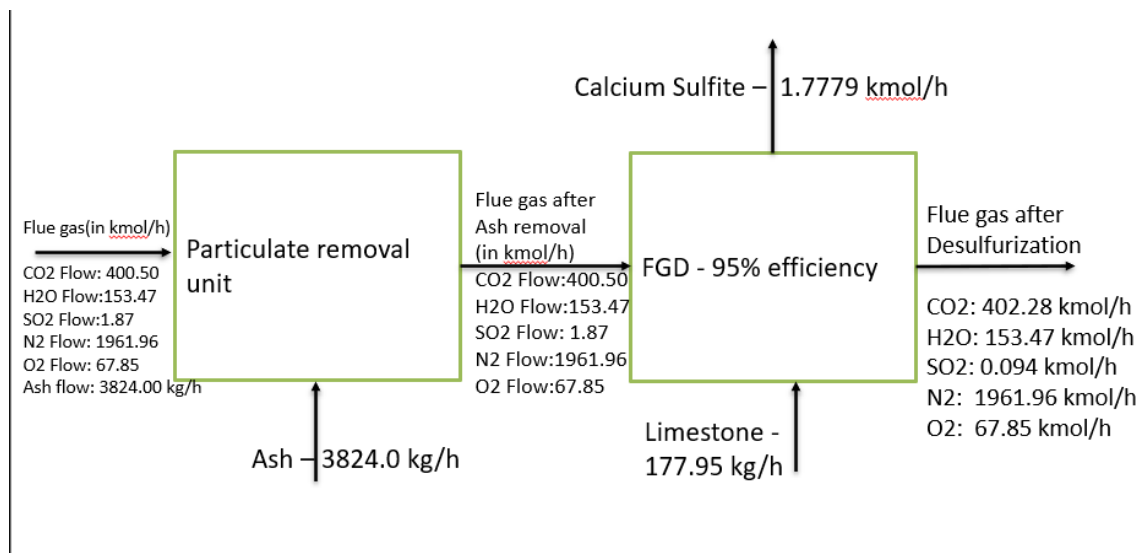


The variables to calculate about each process unit are shown below in the diagram presented for each(or grouped process)unit:-

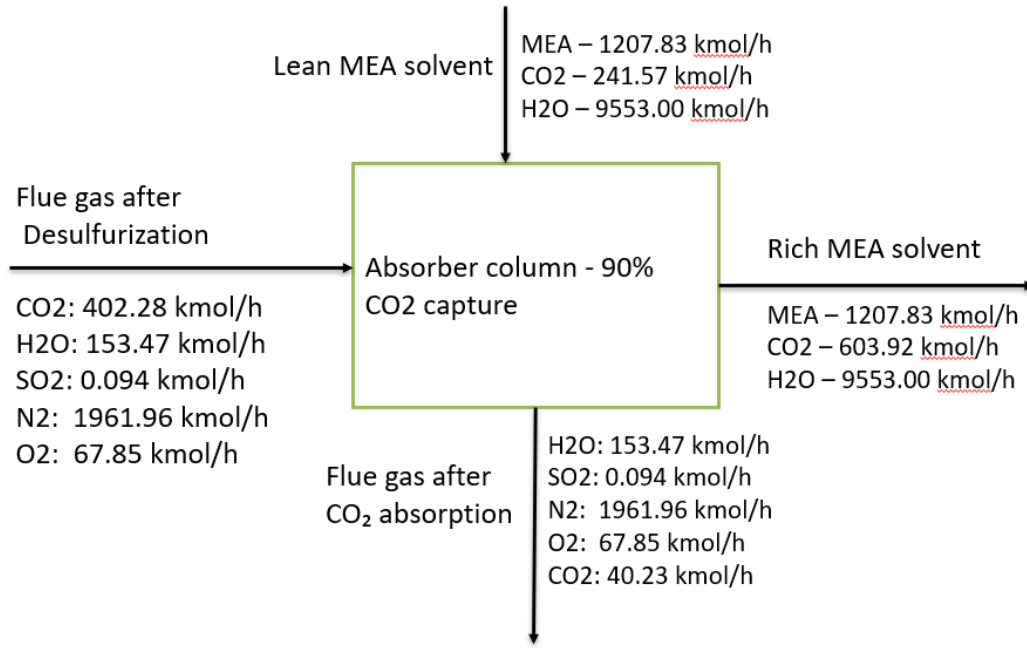
COMBUSTION CHAMBER :-



PARTICULATE REMOVAL UNIT AND FGD :-



ABSORBER COLUMN:-



We have focused on the critical task of dimensioning the absorber column, the core unit for CO₂ capture. This involved calculating both its diameter and height.

We determined the column's diameter using fluid mechanics principles, ensuring stable operation by preventing flooding and minimizing pressure drop, which directly impacts the system's parasitic energy load. For the height, we applied advanced mass transfer concepts, leveraging simplified empirical formulae to ensure the target 90% CO₂ capture efficiency. Additionally, we analyzed how variations in these dimensions affect the absorber's performance, costs, and operational efficiency, providing key insights for optimal design.

Some Key Results are described in the table below :

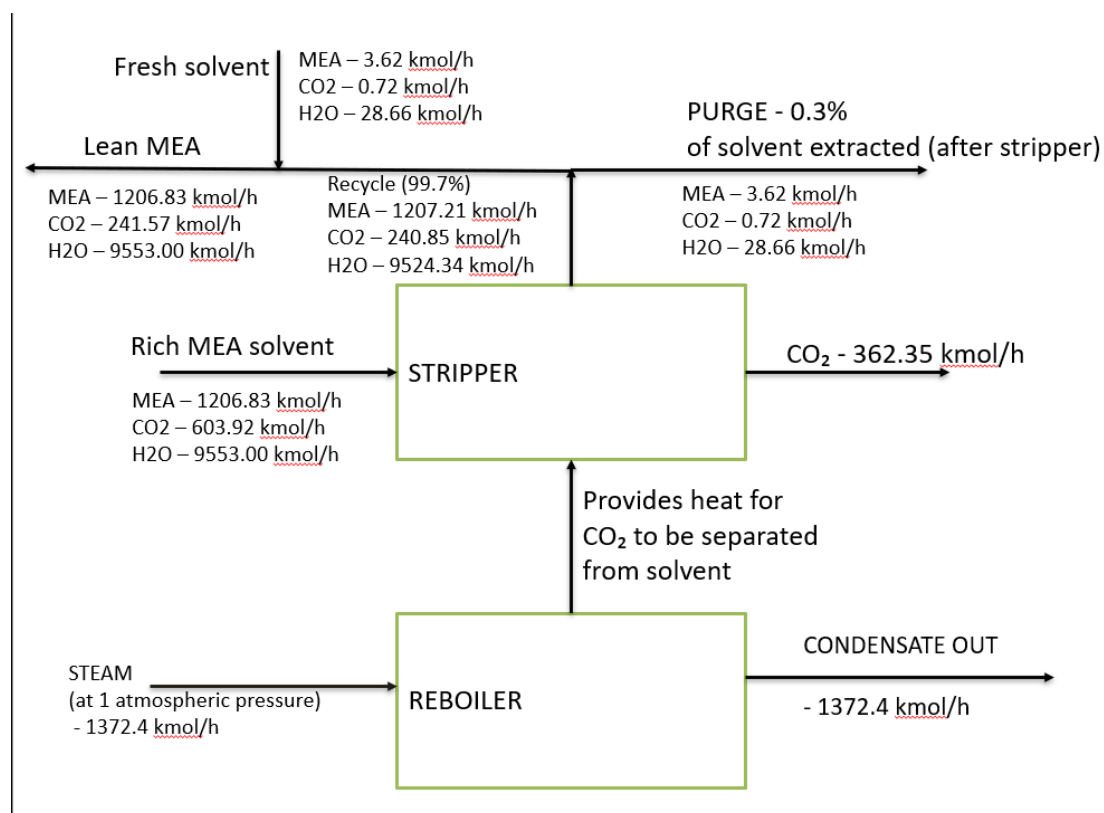
Parameter	Symbol	Value
Column diameter	D_c	4.92 m
Column height	Z	28.5 m
Solving type	–	–
Solvent	–	30 wt% MEA
Capture efficiency	η_{CO_0}	90%
Operating velocity	U_{op}	1.0 m/s

STRIPPER AND REBOILER-

The rich MEA solvent from the absorber, containing about **0.5 mol CO per mol MEA**, is fed to the **stripper column** for regeneration. In this unit, heat is supplied from the **reboiler** at the bottom, which raises the temperature of the rich solvent to approximately **120°C** and releases the absorbed CO₂ gas.

The regenerated lean solvent, having a loading of **0.2 mol CO/mol MEA**, is recycled back to the absorber for reuse. A small **purge of 0.3%** of the circulating solvent (**737 kg/h of 30 wt% MEA**) is extracted after the stripper to control solvent degradation and remove heat-stable salts. The **CO₂ product stream** from the stripper top corresponds to **362 kmol/h (15.9 t/h)** of CO₂ released, which is later cooled and dried to obtain 98% pure CO gas.

The **reboiler** provides the energy required for this regeneration step, with a total heat duty of **1.27 GJ/h**, equivalent to **561 kg/h of saturated steam at 1 atm**. Most of this steam condenses within the reboiler, leaving as **condensate of equal mass flow**, which is usually recovered as boiler feedwater. The combination of the stripper and reboiler enables continuous solvent regeneration and cyclic operation of the CO₂ capture process.



Note-The values presented in the assumptions and schematic are for solving a particular case and as presented in proposal we will have comparisons for different solvents and

would also see the variation pertaining to varying different values.

ANALYSIS

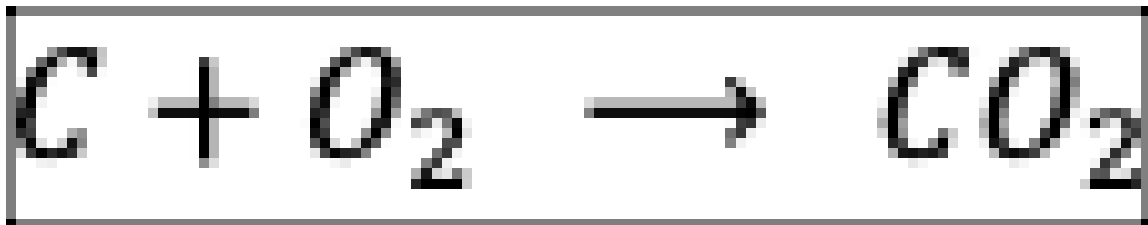
1. Degree of Freedom Analysis

All the equations were linear and they had only one variable hence the calculation was basic and the degree of freedom was 0 for every equation hence resulting in the overall degree of freedom to be 0 for the complete problem

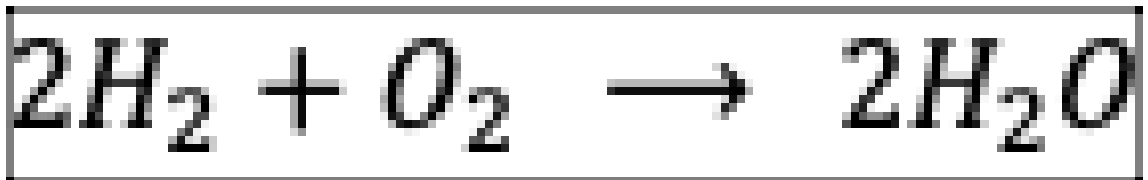
2. Reactions involved

a) Combustion of Coal Elements

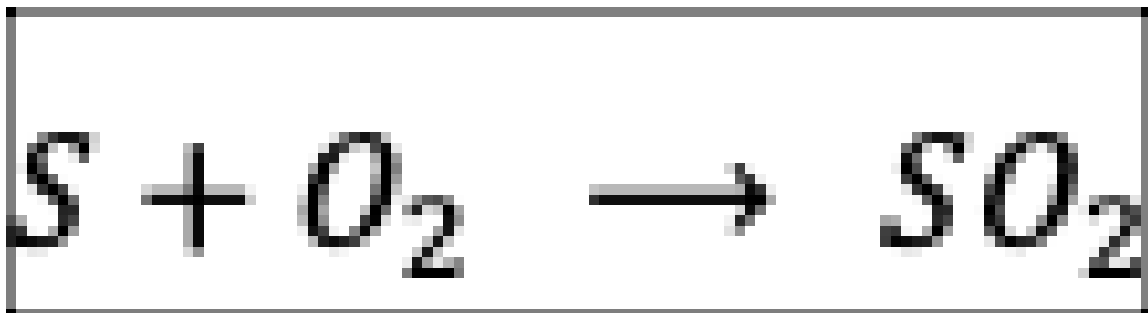
Carbon oxidation:



Hydrogen oxidation:

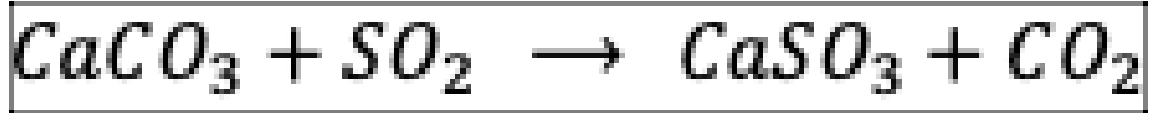


Sulphur oxidation:



b) Flue Gas Desulfurization (SO Capture by Limestone)

Reaction forming calcium sulphite:



c) CO₂ Absorption in MEA (Absorber)

The overall reaction of CO with monoethanolamine (MEA) is:

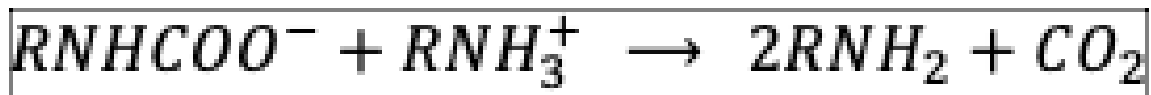
Carbamate formation:



where NH₂ represents MEA.

d) CO₂ Desorption in Stripper (Regeneration)

Carbamate decomposition:



Results and Discussion

Solvent Analysis for CO₂ Absorption

A comparative study of Monoethanolamine(MEA), Methyldiethanolamine(MDEA) and sodium hydroxide(NaOH) highlights MEA as the most practical solvent for large scale CO₂ capture. MEA is a primary amine, reacts rapidly with CO₂ to form carbamates, ensuring high absorption efficiency and reliable regeneration, though at a higher energy cost. MDEA, a tertiary amine, offers lower corrosion and energy requirements but slower reaction kinetics, often requiring blending to match MEA's performance. NaOH, while highly reactive and inexpensive, forms non-regenerable carbonates leading to excessive

waste and unsuitability for continuous operations. Balancing kinetics, cost, regeneration and industrial feasibility, MEA remains the industry benchmark for efficient and economical CO₂ removal.

Analysis of Solvent Purge and Management

The **purge stream** is essential for maintaining the **chemical stability and optimal performance** of the MEA solvent over the long term. Continuous solvent reuse is unsustainable due to the accumulation of **non-volatile contaminants**, and the purge is taken from the lean solvent stream after the stripper.

Necessity and Consequences

Contaminants primarily include **Heat Stable Salts (HSS)**, formed by the irreversible reaction of residual SO₂ (which bypasses the 95% efficient FGD unit) with MEA, and **Solvent Degradation Products** from thermal/oxidative breakdown in the stripper/reboiler. The contaminant generation rate is defined by the **0.60wt% coal sulfur content** and the FGD performance.

Failure to purge leads to severe operational penalties, including:

Reduced CO₂ Capacity: HSS bind to the MEA, reducing the active solvent concentration.

Increased Energy Demand: Contaminated solvent requires more steam, increasing reboiler duty (3.5 MJ/kg CO₂).

Corrosion: HSS significantly increase the corrosivity of the solvent.

Optimal Purge Rate

The **Optimal Purge Rate** (Q_{Optimal}) is calculated to minimize the **Total Annual Operating Cost (TAC)**, which represents the critical trade-off between the cost of **Fresh Solvent Makeup** and the **Cost Penalty from Contamination**.

$$\text{TAC} = (\text{Cost of Fresh Solvent Makeup}) + (\text{Cost Penalty from Contamination})$$

Coal Feed Basis: 10000 kg/h

Molar Feed (C): 400.50 kmol/h

Molar Feed (H): 306.93 kmol/h

Molar Feed (S): 1.87 kmol/h

Molar Feed (N₂): 5.00 kmol/h

Molar Feed (O₂): 26.75 kmol/h

Mass Feed (Ash): 3824.00 kg/h

Theoretical O₂ Required from Air: 452.35 kmol/h

Stoichiometric Flue Gas Flow: 2260.05 kmol/h

Actual O₂ Supplied: 520.21 kmol/h

Actual N₂ Supplied: 1956.97 kmol/h

Unreacted (Excess) O₂: 67.85 kmol/h

Raw Flue Gas (pre-FGD)

CO₂ Flow: 400.50 kmol/h

H₂O Flow: 153.47 kmol/h

SO₂ Flow: 1.87 kmol/h

N₂ Flow: 1961.96 kmol/h

O₂ Flow: 67.85 kmol/h

Ash flow: 3824.00 kg/h

Total Gas Flow: 2585.65 kmol/h

Limestone (CaCO₃) Required: 177.95 kg/h

Gas to Absorber (FG-ASH-SO₂)

CO₂: 402.28 kmol/h

H₂O: 153.47 kmol/h

SO₂: 0.094 kmol/h

N₂: 1961.96 kmol/h

O₂: 67.85 kmol/h

Total Flow: 2585.65 kmol/h

Calculating Solvent Circulation

CO₂ Captured: 362.05 kmol/h (15933.81 kg/h)

Required MEA Molar Flow: 1206.83 kmol/h

Total Solvent Circulation Rate (30% MEA): 245711.10 kg/h

Calculating Reboiler Duty

Total Reboiler Duty: 55768.33 MJ/h (15.49 MW)

Steam Required (at 2257 kJ/kg): 24709.05 kg/h

Carbon Credits, Sustainability, and Strategic Value

Carbon credits, defined as tradable permits allowing the emission of 1 ton of CO₂, are driving a mandatory corporate transition toward carbon-neutral operations and financial accountability. Companies now view carbon capture as a strategic investment that aligns environmental goals with long-term business resilience. Though it involves higher short-term costs, it strengthens competitiveness, attracts ESG-linked finance, and mitigates carbon-tax risks. By supporting UN SDGs—notably Clean Energy, Industry Innovation, Responsible Production, and Climate Action—corporations can turn climate responsibility into a source of innovation, investor trust, and lasting value creation, shifting sustainability from a regulatory burden to a strategic growth driver.

Conclusions

The project was a success in terms of calculation for flue gas composition and absorber sizing and design parameters calculation also the analysis for cases was also done. The calculation was done on matlab. The analysis involved purge importance difference between MEA, MDEA, NAOH and also covered sustainability angles and relevance of carbon credits.